ISLAMIC UNIVERSITY OF TECHNOLOGY
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Final Project Report

Airport Management System

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Team Members

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SPL-I

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Problem Statement

Airport operations involve multiple interdependent activities such as flight scheduling, gate and runway allocation, passenger booking, check-in, and boarding. When these activities are handled manually or through disconnected processes, they often result in scheduling conflicts, inefficient use of resources, delays in operational decisions, and poor passenger service during disruptions. In addition, weather-related changes and limited infrastructure make airport coordination more difficult. Therefore, there is a need for a centralized system that can manage airport workflows in an organized, secure, and efficient manner.

Potential Solution

The proposed solution is a simulation-driven Airport Management System that integrates the core activities of a small airport into a single terminal-based application. The system is designed to automate flight scheduling, passenger booking and check-in, gate and runway allocation, role-based access control, and weather-aware operational decisions. By bringing these modules together in one platform, the system reduces manual dependency, improves coordination among airport functions, and creates a more reliable operational workflow.

Implemented Features

The developed Airport Management System is a Command-Line Interface (CLI)-based application that integrates multiple core functionalities to ensure smooth, efficient, and realistic airport operations under both normal and adverse conditions. The system supports comprehensive flight management, where all flight-related data such as schedules, status (on-time, delayed, holding, cancelled), passenger count, and operational constraints are maintained in an organized

manner. Resource management is implemented to efficiently allocate critical infrastructure such as gates and runways, minimizing conflicts and ensuring optimal utilization.

The system provides flexible user access through the CLI, where users can explore flights as guests without logging in, while passengers can sign up and sign in to access booking features. Registered passengers can select flights, view available and occupied seats separately, perform full or partial seat bookings and cancellations, and receive booking confirmations. The system also supports conditional refund policies, where cancellations made more than six hours before departure are eligible for partial refunds based on a percentage of the ticket price. Administrative access is handled through a dedicated admin login, through which all operational management tasks such as flight control, scheduling, and system monitoring are performed.

A simulation engine is integrated within the CLI environment to mimic real-world airport operations, enabling key actions such as boarding preparation, arrival processing, and dynamic flight rescheduling. The engine continuously attempts to manage and recover delayed operations while generating detailed logs and summaries. These logs provide real-time insights into flight statuses (active, delayed, holding, cancelled), passenger activities (booked, cancelled, departed), and overall system behavior during both manual and automated simulation runs.

A major strength of the system lies in its advanced weather management and disruption handling. During adverse weather conditions such as storms, the system performs full rescheduling using batch scheduling, where each batch contains an equal number of flights relative to available resources. For arrival or holding flights, the next batch begins after approximately 51 minutes, considering 30 minutes for arrival processing and 20 minutes for resource release. For departure or delayed flights, batch intervals extend to around 81 minutes, accounting for 60 minutes of boarding-to-departure time and an additional 20-minute buffer.

The system also handles unexpected operational chaos caused by technical issues or missed time windows. If 1–3 flights miss their processing or boarding windows, they are managed using a single resource through mini batch scheduling. If more than three flights are affected, the system triggers a full rescheduling of all active

flights that have not yet been allocated resources, utilizing all available resources. During scheduling and recovery, strict priority is maintained: holding flights (arrivals that missed processing and are still in the air) are handled first, followed by currently active flights, and then delayed departure flights. Flights exceeding their allowable operational duration are automatically marked as diverted (for arrivals) or cancelled (for departures).

Before assigning resources to any flight, the system ensures that previously occupied resources are freed and reallocated efficiently to flights that are still within their valid operational time window. The system uses file-based persistence to store flight data, passenger information, and login credentials across sessions. It also generates structured text-based reports for monitoring and analysis.

Overall, the system delivers a robust, adaptive, and realistic airport operation model within a CLI environment by combining intelligent scheduling, simulation capabilities, passenger services, and dynamic disruption management into a unified platform.

Tools and Technology

Programming Language: Java
(The core programming language used to develop the system, implementing object-oriented design for better modularity and maintainability.)

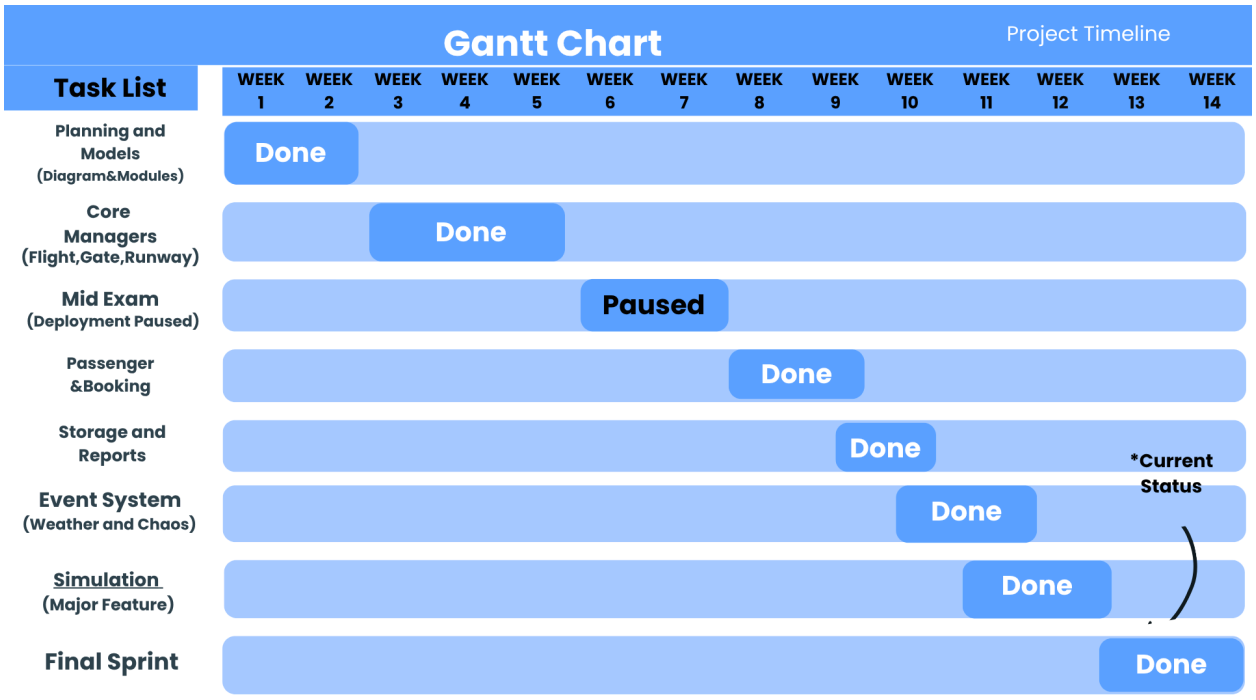
Development Environment: IntelliJ IDEA
(The Integrated Development Environment (IDE) used for coding, debugging, and ensuring clean architecture in the project.)

Version Control: GitHub
(A platform for managing code versions, facilitating collaboration, and tracking changes across the development process.)

Application Domain

This project belongs to the domain of **airport operations management** and **transportation system software**. More specifically, it is suitable for small-airport simulation, academic demonstration of resource allocation and workflow automation, and training-oriented environments where flight operations, passenger handling, and infrastructure coordination must be represented in a simplified but functional manner.

Project Timeline



Contribution

230042115-Sheikh Alimun Alahi's Contributions

- **Gate and Runway Management System**
  (Developed the system responsible for managing the allocation and status of gates and runways.)
- **Seat Layout**
  (Designed the seating arrangement for flights, ensuring proper seat assignments and capacity handling.)
- **Console Layout**
  (Created the user interface layout for the command-line terminal to improve user experience.)
- **Weather Type and Weather Slot Features**
  (Implemented weather type selection and slot management, which influence flight scheduling and operational decisions.)
- **Simulation Engine**
  (Developed the engine driving the simulation, including time updates, weather changes, and event handling.)

230042128-Rakibur Rahman Ananta's Contributions

- **Passenger Management**
  (Developed the system for storing and managing passenger data, ensuring effective check-ins and boarding.)
- **App User and Authentication**
  (Implemented user management and authentication systems to ensure secure access to various modules.)
- **Password Security Utilities**
  (Created password encryption and validation mechanisms to ensure secure user login and data integrity.)

230042147-Irfan Sadik Chy's Contributions

- **Flight and Flight Management**
  (Designed and managed the system for flight scheduling, status tracking, and operational conditions.)
- **Departure and Arrival Handling**
  (Implemented the systems that handle flight departures and arrivals, ensuring accurate scheduling and monitoring.)
- **Booking Management**
  (Developed the booking system that handles seat selection, ticket generation, and booking confirmations.)
- **Weather Manager**
  (Created the module for managing real-time weather conditions that impact flight clearance and scheduling.)

Future Work :

The project can be extended in several important directions. First, it can be integrated with **real-time airline APIs** to support live operational data. Second, a **mobile application for passengers** can be developed to improve accessibility and user interaction. Third, an **AI-based delay prediction system** can be introduced to improve forecasting and operational planning. In addition, since the current system uses file-based storage, future versions may be enhanced with a **database-backed architecture** to improve scalability, data consistency, and real-world applicability.